B.TECH PROJECT · PROBLEM STATEMENT

Optimising Phase Change Material (PCM) Retrofits on the Roof and Walls of PMAY Houses for Thermal Comfort and Energy Savings in a Warm-Humid Climate

Indian Institute of Technology Kharagpur · Study location: Kolkata / Kharagpur, West Bengal (warm-humid climate) · Version 1.1, 22 October 2026

1. Background and motivation

Space cooling is the fastest-growing use of energy in Indian buildings. It already accounts for about 38 % of building-sector energy use and is projected to reach 50 % by 2027 [1]. India experiences roughly 28 % of the world's population-weighted cooling degree-days, yet fewer than 7 % of homes own a room air conditioner [2]. As incomes rise this gap will close quickly, with large consequences for peak electricity demand and emissions. Every passive measure that keeps homes comfortable without air conditioning, or reduces how hard an air conditioner must work, therefore matters at national scale.

Pradhan Mantri Awas Yojana – Urban (PMAY-U) is delivering millions of small, permanent houses to economically weaker households. The West Bengal model design for *Beneficiary-Led Construction* (BLC), shown in Fig. 1, is typical: a 25.8 m² carpet-area unit with 250 mm burnt-brick walls and an exposed, uninsulated reinforced-cement-concrete (RCC) roof slab [12]. In a warm-humid climate such a roof absorbs intense solar radiation through the day and keeps radiating that heat into the rooms well into the night. The occupants, who mostly cannot afford air conditioning, bear the consequences.

Phase change materials (PCMs) store a large amount of heat at an almost constant temperature while they melt, and give it back when they solidify. Placed in the building envelope, a PCM layer can absorb part of the daytime heat flow, delaying and flattening indoor temperature peaks. In Rupnagar (Punjab), an RCC roof slab with macro-encapsulated PCM cut the heat flux into the room by 60.6 % and the thermal load by 49.8 %, with a simple payback of 5.7 years [1]. Independent PCM ceiling panels have even held a lightweight enclosure below the outdoor temperature [3, 4]. Because a PCM layer can be added to an existing house (on top of the roof slab, as a ceiling board, or as an inner wall lining) without rebuilding it, PCM is a candidate *retrofit* for the large stock of PMAY houses already built.

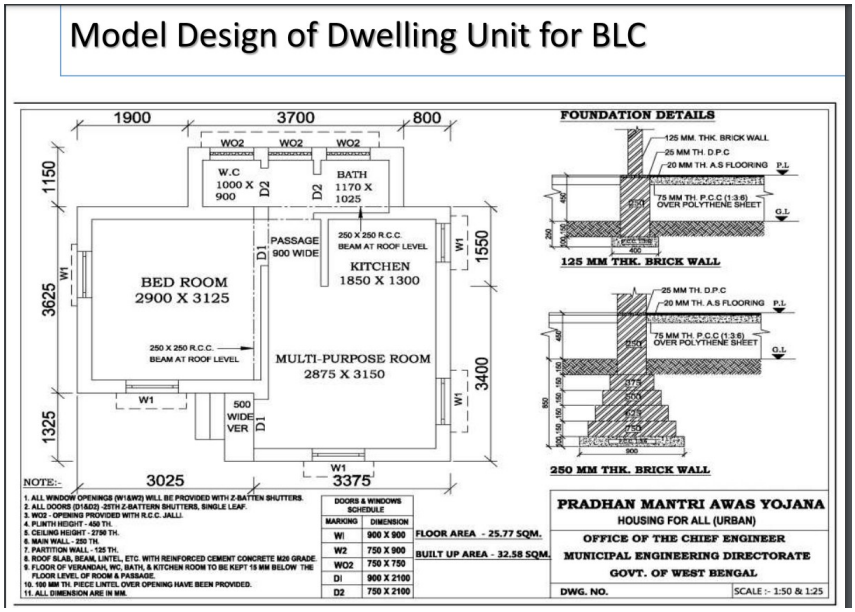


Fig. 1. Reference building: PMAY-U model design of a dwelling unit for BLC, Municipal Engineering Directorate, Government of West Bengal [12]. Floor area 25.77 m², built-up area 32.58 m².

2. Why this is not straightforward

A PCM only helps if it completes a melt–freeze cycle every day: it must melt while absorbing daytime heat and then *re-solidify at night* to be ready for the next day. Fig. 2 shows why this is the central difficulty in Kolkata. From April to September the mean night-time temperature is 28.2 °C, and on 58–94 % of nights the outdoor minimum stays above 26 °C. n-Octadecane, the PCM proposed for this project, melts at about 28 °C, so it may stay partly molten for weeks and stop working exactly when it is needed. The opposite problem appears in winter (December–February, mean night temperature 17.8 °C), when a 28 °C PCM placed indoors may hardly melt at all. On the outer face of a sun-baked roof, temperatures far above 50 °C change the picture again. In Chennai, a similar climate, a single PCM layer in a roof did not complete its melt–freeze cycle [15].

The PCM also has to compete with simpler measures. Eco-Niwas Samhita 2018, India's residential energy code, limits the roof U-value to 1.2 W/m²K in warm-humid climates [18]. The bare RCC roof of the PMAY house (about 3.4 W/m²K) is almost three times worse. A thin PCM layer adds latent storage but little insulation, so it must be judged against, and in combination with, cheap alternatives such as roof insulation and reflective 'cool-roof' coatings. Work in warm-humid Indian offices found PCM alone inadequate and an insulation–PCM combination best [16].

The best PCM, its melting temperature, thickness and position therefore depend on the climate, the orientation of each surface and on how the house is operated (windows, ceiling fans, air conditioning). These choices must also be balanced against cost for a house built on a very small budget. Finally, comfort has to be judged with the right yardstick. The India Model for Adaptive Comfort – Residential (IMAC-R), derived from year-long surveys in Indian homes including Kolkata, shows that the PMV model under-predicts how much warmth Indian occupants accept [2]. IMAC-R is therefore the appropriate measure for naturally ventilated PMAY houses.

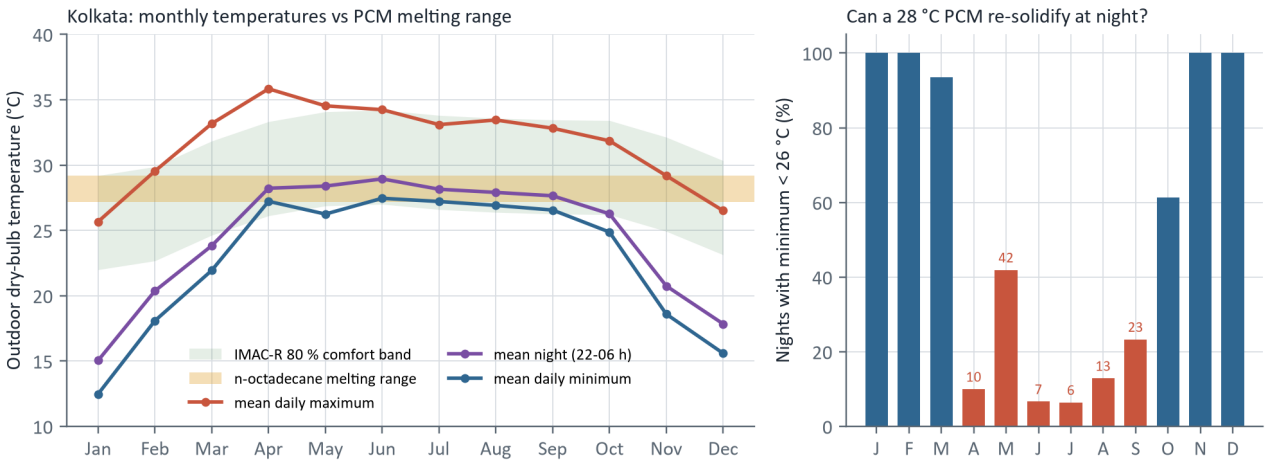


Fig. 2. Left: Kolkata monthly mean maximum, night-time (22–06 h) and minimum outdoor temperature (TMYx 2011–2025 [13]) against the melting range of n-octadecane and the IMAC-R 80 % comfort band. Right: share of nights on which the outdoor minimum falls below 26 °C, i.e. when a 28 °C PCM could fully re-solidify by exchanging heat with outdoor air.

3. Problem statement

Problem statement

To identify, through verified building-energy simulation and multi-objective optimisation, the PCM retrofit (choice of PCM and its melting temperature, layer thickness, and location on the roof and walls) that gives the best trade-off between **occupant thermal comfort** (IMAC-R), **cooling energy use** (Energy Performance Index) and **investment cost** for the West Bengal PMAY-U BLC dwelling in the warm-humid climate of Kolkata/Kharagpur, under passive (free-running), ceiling-fan-assisted and air-conditioned operation.

Research gaps addressed

The gaps below are drawn from a review of 47 verified sources (work/literature/literature_review.md).

- **Climate and building type.** Experimental PCM-roof work in India has concentrated on hot, dry or composite climates and on small test units [1]. The only peer-reviewed simulation study of PMAY houses examined wall assemblies against the energy code, not PCM [17]. Whole-house PCM assessments of low-cost dwellings in the warm-humid east of India are scarce.
- **Thermal mass.** Much recent PCM-panel research targets lightweight, low-mass enclosures [3, 4]. Brick-and-RCC houses already store heat, so the added benefit of PCM there is less certain.
- **Night-time discharge.** Incomplete re-solidification in climates with warm nights, and how it should drive melting-temperature selection, is seldom quantified for Indian houses.
- **Comfort yardstick.** PCM benefits are usually reported as temperature reductions, PMV or energy savings. India's residential adaptive model (IMAC-R) [2] is seldom used to judge them.
- **Joint optimisation.** Placement (roof vs. individual walls), melting temperature, operating mode and cost have not been optimised together for affordable housing. PCM has also seldom been benchmarked against cheap conventional retrofits; envelope measures that look good on paper can even worsen comfort in free-running Indian homes [19].

4. Aim and objectives

Aim: to recommend a practical, cost-effective PCM retrofit for PMAY houses in warm-humid West Bengal.

1. Build and verify an EnergyPlus model of the PMAY BLC house in Kolkata, and establish its baseline comfort and energy performance.

2. Model PCM layers on the roof (top of slab or ceiling underside) and on each wall, with n-octadecane as the baseline PCM and n-alkanes melting between 22 °C and 37 °C as alternatives. Verify the PCM model (grid and time-step independence, and a control case without latent heat).
3. Quantify, for passive, fan and air-conditioned operation, the effect of each configuration on IMAC-R discomfort hours, peak temperatures and cooling energy, separately for summer, monsoon and winter.
4. Quantify how often the PCM actually completes its melt–freeze cycle, and identify suitable melting temperatures for the warm-humid climate.
5. Find Pareto-optimal retrofits for comfort, energy and cost with NSGA-II/NSGA-III, and compute payback and CO_2 savings.
6. Translate the results into retrofit guidelines for PMAY houses.

5. Scope and study design

5.1 Reference building and simulation model

Item	Value / assumption
Location & climate	Kolkata (22.65° N, 88.45° E), warm-humid climate zone (NBC 2016 [7]). Kharagpur checked with Midnapore weather (WMO 428030, ~10 km away).
Weather data	TMYx 2011–2025 typical-year files from climate.onebuilding.org [13]
Geometry	Bedroom (BED), multi-purpose room + kitchen + passage (LIV), WC + bath (TOI); 2.75 m ceiling; 31.9 m² modelled floor area (Fig. 3)
Walls	15 mm cement plaster + 250 mm burnt brick + 12 mm plaster (U ≈ 2.0 W/m²K); 125 mm brick partitions
Roof	25 mm cement screed + 125 mm M20 RCC slab + 12 mm plaster (U ≈ 3.4 W/m²K); slab thickness assumed (not on drawing)
Openings	5 windows W1 (0.9 × 0.9 m, single clear glass), entrance door, 3 jali ventilators to the toilet block
Occupancy	Family of four; bedroom used 21–07 h, living room all day; lights 3 W/m², fridge, TV, LPG cooking (1 kW, 2.5 h/day)
Ventilation	Windows open 06–22 h when the room is warmer than outdoors and warmer than the day's IMAC-R 90 % lower comfort limit; night ventilation is a study variable; infiltration 1 ACH
Orientation	Not given on the drawing: plan-up = North as the base case, with rotation as a sensitivity check

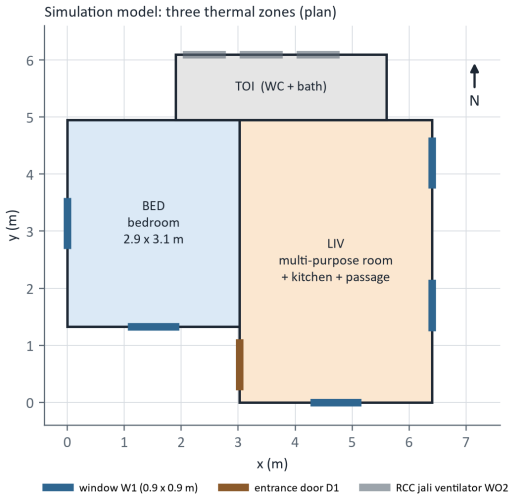


Fig. 3. Plan of the simulation model derived from Fig. 1 (centre-line dimensions). PCM retrofits are applied only to the habitable rooms (BED, LIV).

5.2 Configurations (from the proposed layout of work)

Group	Configuration	How it is modelled
Passive	PCM on roof	PCM layer on top of the slab under a new 25 mm screed, *or* as a ceiling lining (PCM + 12.5 mm gypsum board)
Passive	PCM on walls	Inner-face PCM lining on the North, East, South, West or all walls of the habitable rooms (16 combinations)
Passive	Walls + roof	Any wall combination together with either roof position
Mixed	Ceiling fan	Every passive configuration with a 60 W ceiling fan per room. The IMAC-R upper limit is raised by 1.2 °C for elevated air speed (ASHRAE 55 [6])
Mixed	Air conditioning	Every passive configuration with a split AC (26 °C set-point, COP 3.3) in the bedroom at night and in the living room in the afternoon and evening (1 Mar–15 Nov)

5.3 Design variables

Variable	Levels in the parametric study	In optimisation
Roof PCM position	none / top of slab / ceiling underside	categorical
Wall PCM	any subset of N, E, S, W (16)	4 binary
PCM (melting point)	n-heptadecane 22 °C, **n-octadecane 28 °C** (baseline), n-nonadecane 32 °C, n-eicosane 36.6 °C [8]	continuous 20–38 °C
Layer thickness	5, 10, 20 mm	continuous 3–30 mm
Night ventilation	windows closed / open at night	binary
Orientation	0° (base); 90°, 180°, 270° as sensitivity	fixed
Benchmark measures	cool-roof coating; XPS roof insulation 25 / 50 mm; combinations with PCM	XPS 0–50 mm, cool roof yes/no

6. Methodology

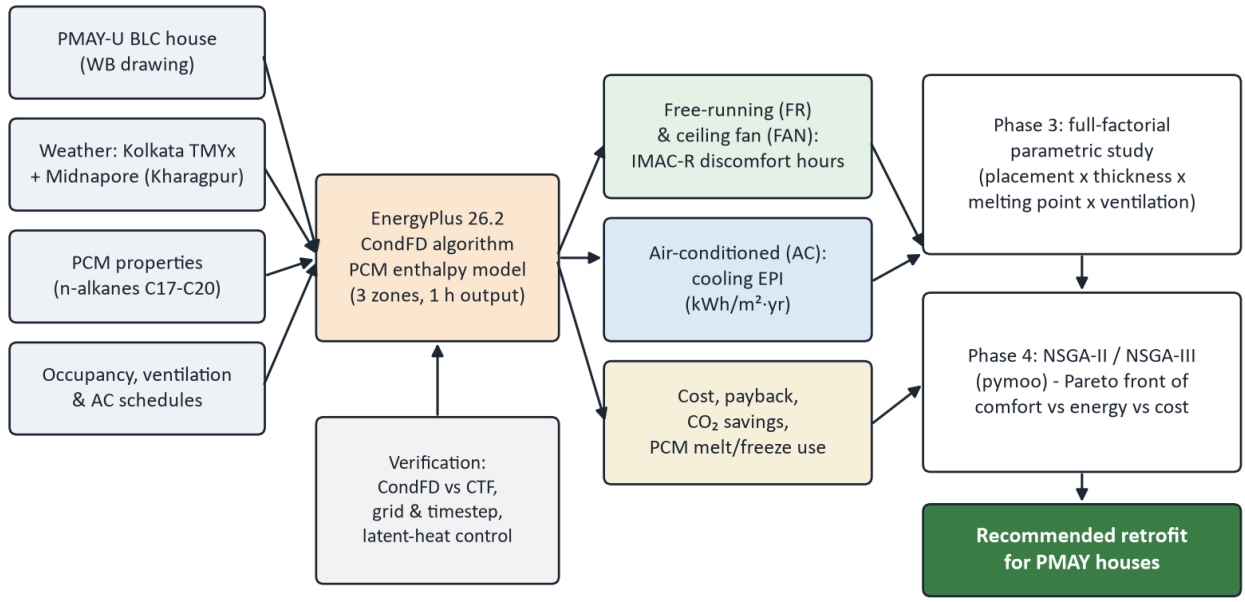


Fig. 4. Methodology and work-flow.

- **Simulation engine:** EnergyPlus 26.2 [9] with the Conduction Finite Difference (CondFD) algorithm, which represents the PCM by its enthalpy–temperature curve and temperature-dependent conductivity. A 3-minute time step is used, and every case (including the no-PCM baseline) uses the same algorithm so

comparisons are fair.

- **Model generation and automation:** the model is written by a Python script (geometry from the drawing, constructions, schedules), so thousands of variants can be generated and run in parallel without manual editing.
- **Verification:** CondFD against the standard conduction-transfer-function solver for the baseline; sensitivity to the grid and time step; a control case with the same layer but no latent heat, to separate the PCM effect from the extra layer's resistance and mass; and checks on the PCM melt–freeze state [14]. No measured data exist yet for this house, so short-term indoor temperature logging in a real PMAY house near Kharagpur is proposed for validation.
- **Optimisation:** NSGA-II [10] and NSGA-III [11] (pymoo). Optimisation is first run on the discrete parametric space, where the true Pareto front is known from brute force, to check that the optimiser finds it. It is then run on an extended continuous space (melting point, thickness).

Performance indicators

Indicator	Definition
Discomfort hours	Occupied hours with indoor operative temperature outside the IMAC-R 80 % band: $T_n = 0.42 \cdot T_{out,30DRM} + 17.6$ °C, band ±3.6 °C (±2.15 °C for 90 %) [2]
Discomfort degree-hours	Sum of the temperature excess above (or below) the band over occupied hours
Peak temperature & swing	Maximum operative temperature; mean daily indoor swing in summer
Cooling EPI	Annual AC electricity per m² of floor area (kWh/m²·yr)
Seasonal performance	Winter (Dec–Feb), summer (Mar–May), monsoon (Jun–Sep), post-monsoon (Oct–Nov)
PCM utilisation	Share of days with a full melt–freeze cycle; mean daily swing of melt fraction
Cost, payback, CO_2	PCM at Rs 450/kg [1] + encapsulation and finish; payback = cost / (annual kWh saved × Rs 7.5/kWh, WBSEDCL domestic slab); CO_2 = kWh saved × 0.675 kg/kWh (CEA CO_2 Baseline Database v22, FY 2025-26)

7. First results

The baseline model (no PCM, free-running, windows opened by day when the room feels warm) already shows the problem. Indoor operative temperature peaks at 37.3 °C, the ceiling surface reaches 45 °C, and about 90 kWh of heat per m² of roof enters the bedroom each year. 9.7 % of occupied hours fall outside the IMAC-R 80 % band (32 % outside the stricter 90 % band). Hot discomfort is concentrated in pre-monsoon afternoons, and cold discomfort in winter nights.

Early benchmark runs sharpen the question. A 10 mm n-octadecane layer on the roof brings discomfort down to 5.3 % for about Rs 1.2 lakh. A 25 mm XPS insulation layer reaches 1.8 % for about Rs 29,000. A cool-roof coating removes almost all overheating but makes winter colder (30 % of winter hours too cold). The study will therefore establish where, and in which combination, PCM adds value beyond conventional measures.

8. Expected outcomes and deliverables

- A verified, scriptable EnergyPlus model of the PMAY BLC house, and a ranked comparison of all roof and wall PCM configurations for passive, fan and AC operation.
- Evidence on whether n-octadecane suits Kolkata, and which melting temperature works best for each location (roof top, ceiling, walls), including summer vs. winter behaviour.
- Pareto fronts of comfort vs. energy vs. cost, a recommended retrofit with its payback and CO_2 savings, and simple design guidelines.
- A reproducible code repository (GitHub), the BTP-I and BTP-II reports, and (if results permit) a conference or journal manuscript.

9. Work plan

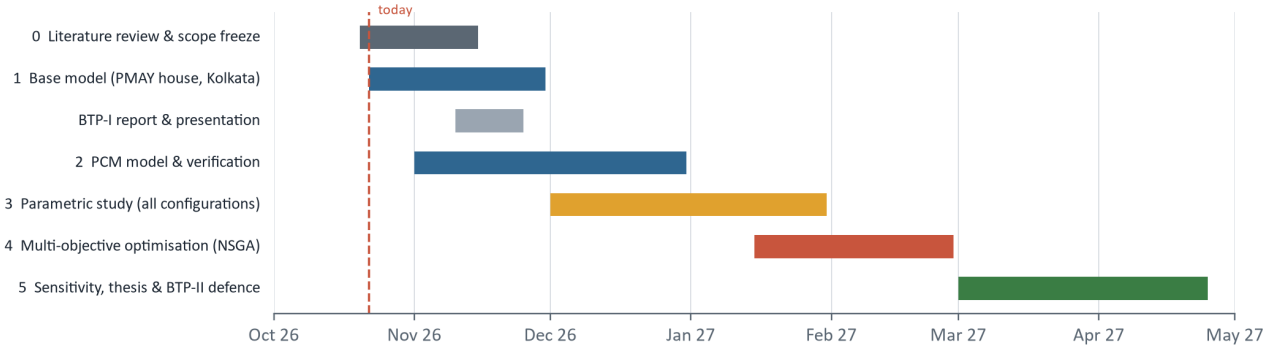


Fig. 5. Work plan for BTP-I (autumn 2026) and BTP-II (spring 2027).

Key assumptions and limitations

- PCM properties are typical literature values for pure n-alkanes [8], and the encapsulation shell is not modelled explicitly. Both will be checked against supplier data.
- Costs, tariff and AC efficiency are indicative and will be varied in a sensitivity analysis.
- A single typical-year weather file is used. Future-climate files may be added if time permits.
- The model is verified numerically. Validation against measured indoor temperatures is planned but depends on access to an occupied PMAY house.

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